



INSTITUTO TECNOLÓGICO
DE LAS AMÉRICAS

ASIGNACIÓN

SEGURIDAD DE REDES

ESTUDIANTE:

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MATRÍCULA:

2025-0827

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TRABAJO:

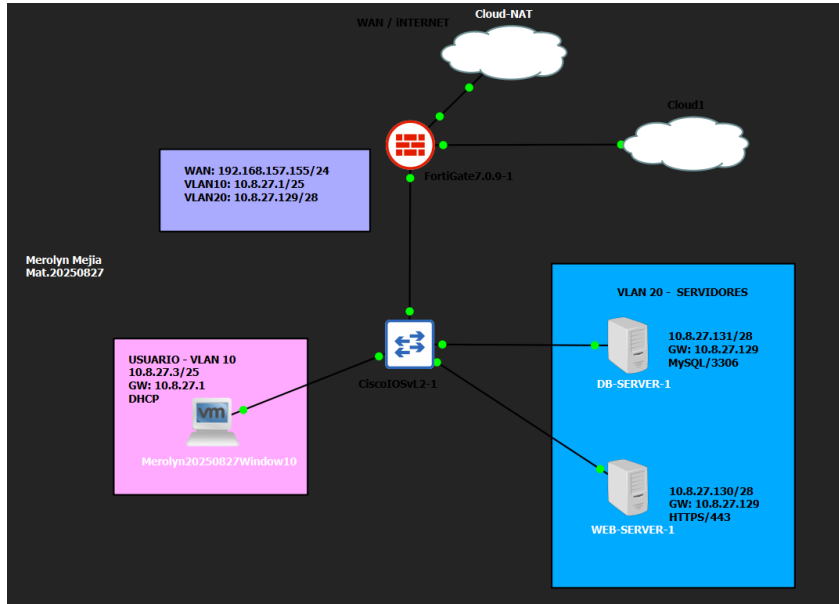
Documentación de Laboratorio

FECHA DE ENTREGA:

25/09/2026



Prueba 01 — Topología general de la red en GNS3



Prueba 02 — Configuración de las VLANs en FortiGate

port2	Physical Interface	0.0.0.0/0.0.0.0	
Servidores (VLAN20-SERVERS)	VLAN	10.8.27.129/255.255.255.240	PING
Usuarios (VLAN10-USUARIOS)	VLAN	10.8.27.1/255.255.255.128	PING HTTPS

Prueba 03 — Asignación de direccionamiento IP al equipo Windows mediante DHCP

```
C:\Users\Merolyn20250827>ipconfig /all

Windows IP Configuration

    Host Name . . . . . : DESKTOP-L9FQTGK
    Primary Dns Suffix . . . . . :
    Node Type . . . . . : Hybrid
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet0 2:

    Connection-specific DNS Suffix . :
    Description . . . . . : Intel(R) PRO/1000 MT Network Connection
    Physical Address. . . . . : 00-0C-29-92-1C-3A
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . : fe80::10bb:9970:69e2:b001%13(Preferred)
    IPv4 Address. . . . . : 10.8.27.3(Preferred)
    Subnet Mask . . . . . : 255.255.255.128
    Lease Obtained. . . . . : Friday, September 25, 2026 10:11:30 AM
    Lease Expires . . . . . : Friday, October 2, 2026 10:11:30 AM
    Default Gateway . . . . . : 10.8.27.1
    DHCP Server . . . . . : 10.8.27.1
    DHCPv6 IAID . . . . . : 352324649
    DHCPv6 Client DUID. . . . . : 00-01-00-01-32-41-3E-6F-00-0C-29-92-1C-3A
    DNS Servers . . . . . : 192.168.157.2
    NetBIOS over Tcpip. . . . . : Enabled

C:\Users\Merolyn20250827>
```

Prueba 04 — Verificación de la ruta por defecto en FortiGate

```
FortiGate-VM64-KVM # get router info routing-table all
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
       O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default

Routing table for VRF=0
S* 0.0.0.0/0 [5/0] via 192.168.157.2, port1, [1/0]
C   10.8.27.0/25 is directly connected, VLAN10-USUARIOS
C   10.8.27.128/28 is directly connected, VLAN20-SERVERS
C   192.168.157.0/24 is directly connected, port1
C   192.168.244.0/24 is directly connected, port3

FortiGate-VM64-KVM #
FortiGate-VM64-KVM #
```

Prueba 05 — Configuración de NAT para salida a Internet

The screenshot shows the FortiGate web interface for editing a Firewall Policy. The left sidebar contains the navigation menu with 'Policy & Objects' expanded and 'Firewall Policy' selected. The main content area displays the configuration for the policy named 'SERVERS-INTERNET'.

Policy Configuration:

- Name:** SERVERS-INTERNET
- Incoming Interface:** Servidores (VLAN20-SERVERS)
- Outgoing Interface:** port1
- Source:** all
- Destination:** all
- Schedule:** always
- Service:** ALL
- Action:** ACCEPT (selected), DENY
- Inspection Mode:** Flow-based (selected), Proxy-based

Firewall / Network Options:

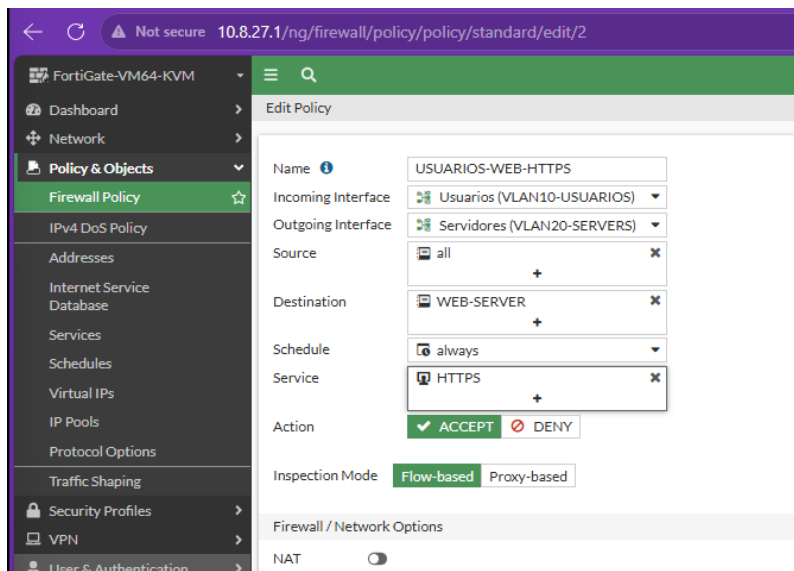
- NAT:** Enabled (toggle switch)

Prueba 06 — Verificación de conectividad hacia Internet

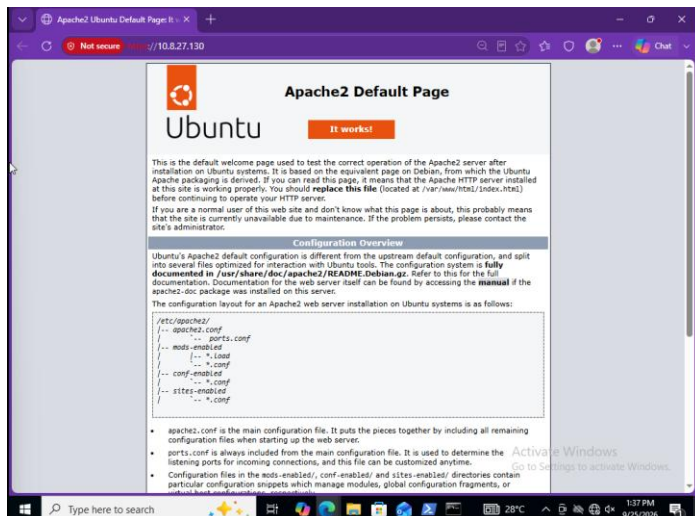
```
ubuntu-server@web-server:~$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=127 time=83.6 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=127 time=74.9 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=127 time=66.9 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=127 time=72.3 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 300ms
rtt min/avg/max/mdev = 66.933/74.450/83.598/6.019 ms
ubuntu-server@web-server:~$
```

Prueba 07 — Política de acceso de Usuarios hacia el WEB Server



Prueba 08 — Acceso al WEB Server mediante HTTPS



Prueba 09 — Verificación de conectividad al WEB Server por el puerto 443

```
PS C:\Windows\system32> Test-NetConnection 10.8.27.130 -Port 443

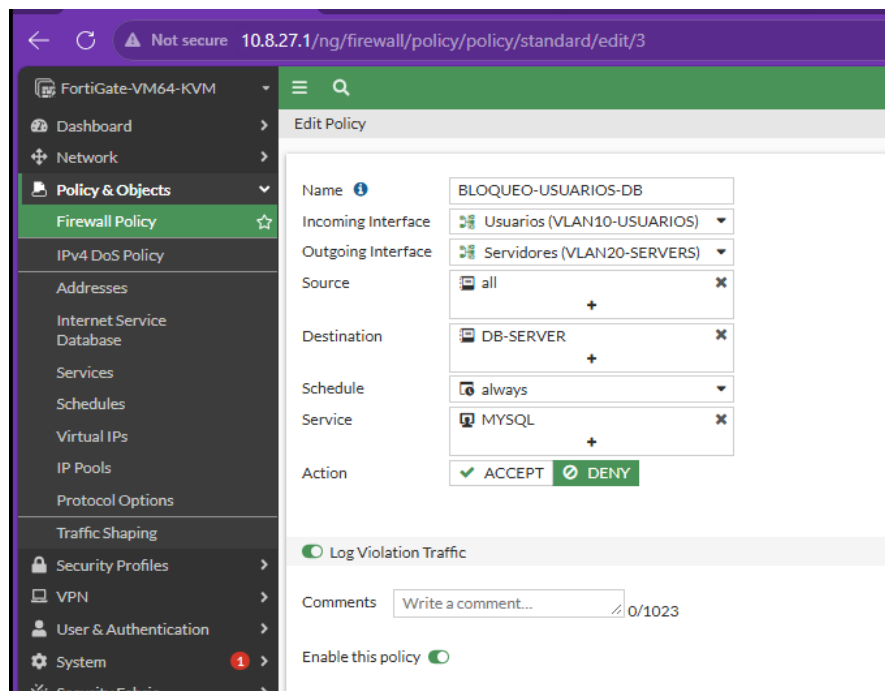
ComputerName      : 10.8.27.130
RemoteAddress     : 10.8.27.130
RemotePort        : 443
InterfaceAlias    : Ethernet0 2
SourceAddress     : 10.8.27.3
TcpTestSucceeded  : True
```

Prueba 10 — Bloqueo de conexión de Usuarios hacia el DB Server por el puerto 3306

```
PS C:\Windows\system32> Test-NetConnection 10.8.27.131 -Port 3306
WARNING: TCP connect to (10.8.27.131 : 3306) failed
WARNING: Ping to 10.8.27.131 failed with status: TimedOut

ComputerName      : 10.8.27.131
RemoteAddress     : 10.8.27.131
RemotePort        : 3306
InterfaceAlias    : Ethernet0 2
SourceAddress     : 10.8.27.3
PingSucceeded     : False
PingReplyDetails (RTT) : 0 ms
TcpTestSucceeded  : False
```

Prueba 11 — Política de bloqueo de Usuarios hacia el DB Serve



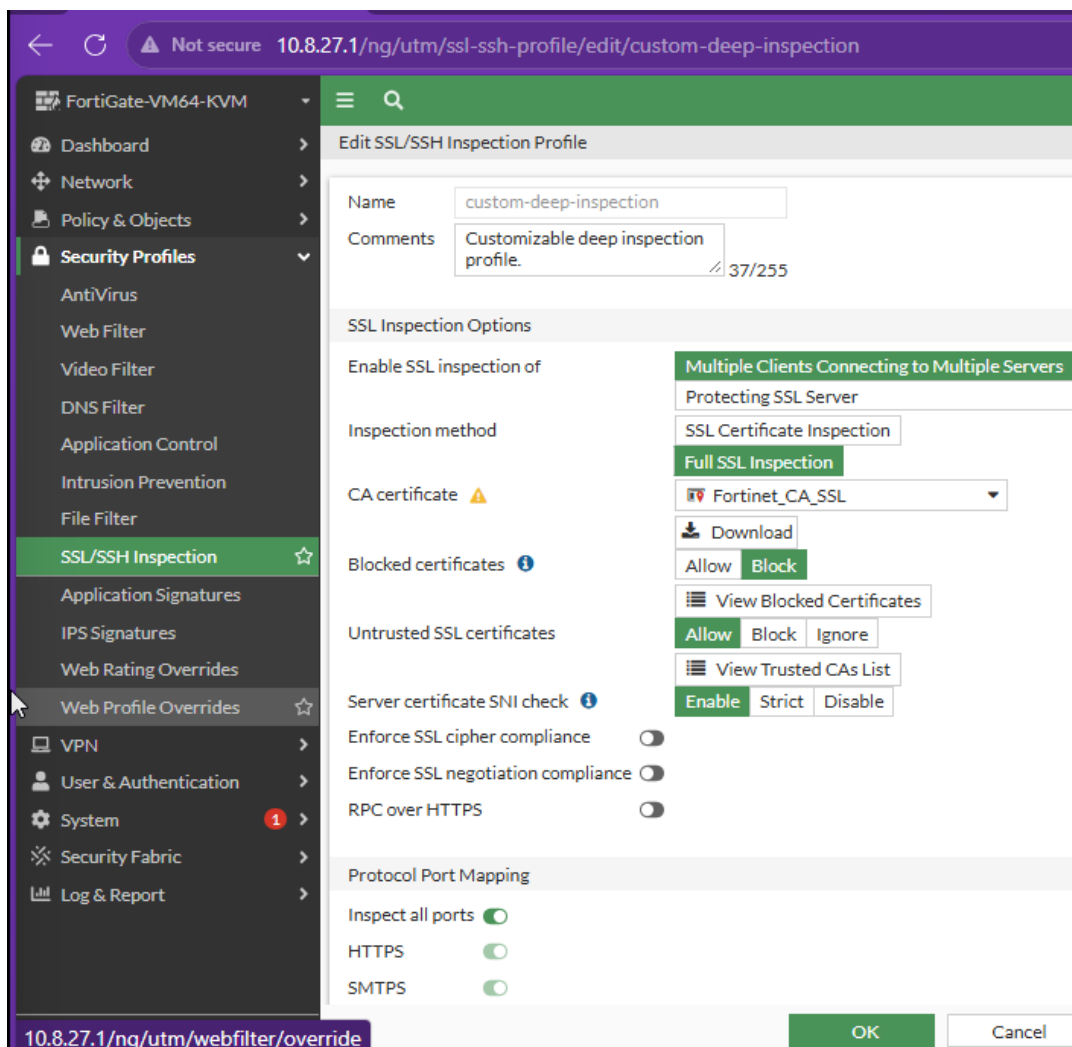
Prueba 12 — Comunicación permitida del WEB Server hacia el DB Server por el puerto 3306

```
ubuntu-server@web-server:~$ nc -zv 10.8.27.131 3306
Connection to 10.8.27.131 3306 port [tcp/mysql] succeeded!
ubuntu-server@web-server:~$ _
```

Prueba 13 — Bloqueo de comunicación por puerto no autorizado entre WEB Server y DB Server

```
ubuntu-server@web-server:~$ nc -zv -w 3 10.8.27.131 80
nc: connect to 10.8.27.131 port 80 (tcp) timed out: Operation now in progress
ubuntu-server@web-server:~$ _
```

Prueba 14 — Configuración de Deep Packet Inspection (DPI)



Prueba 15 — Registro de inspección del tráfico HTTPS mediante DPI

The screenshot shows the FortiGate log viewer interface. The main table displays log entries with columns: Date/Time, Action, Service, Source, Source Interface, and Log Details. The 'Log Details' column is expanded for a selected entry, showing fields like Destination, UUID, Hostname, Application Control, Protocol, Service, Data (Message), Action, Policy ID, Policy UUID, Policy Type, Security Level, Cellular Service, and Other (Log event original timestamp).

Date/Time	Action	Service	Source	Source Interface	Log Details
10 hours ago	Log::Action::resign-as-untru...	SSL	10.8.27.3	Usuarios (VLAN10-USU...	Destination: 72c5ee12-b839-51f1-c0... UUID: 79f26bb37ab1 Hostname: web-server Application Control Protocol: 6 Service: SSL Data: Message: Server certificate is re-signed as untrusted Action: Log::Action::resign-as-untrusted Policy ID: USUARIOS-WEB-HTTPS (2) Policy UUID: e02fe5de-b839-51f1-9c84-9468688d3d3b Policy Type: Firewall Security: Level: [5 bars] Cellular: Service: SSL Other: Log event original timestamp: 1790321633047090200
13 hours ago	Log::Action::resign-as-untru...	SSL	10.8.27.3	Usuarios (VLAN10-USU...	
21 hours ago	Log::Action::resign-as-untru...	SSL	10.8.27.3	Usuarios (VLAN10-USU...	
21 hours ago	Log::Action::resign-as-untru...	SSL	10.8.27.3	Usuarios (VLAN10-USU...	
21 hours ago	Log::Action::resign-as-untru...	SSL	10.8.27.3	Usuarios (VLAN10-USU...	

Prueba 16 — Configuración del perfil IPS para protección contra SQL Injection

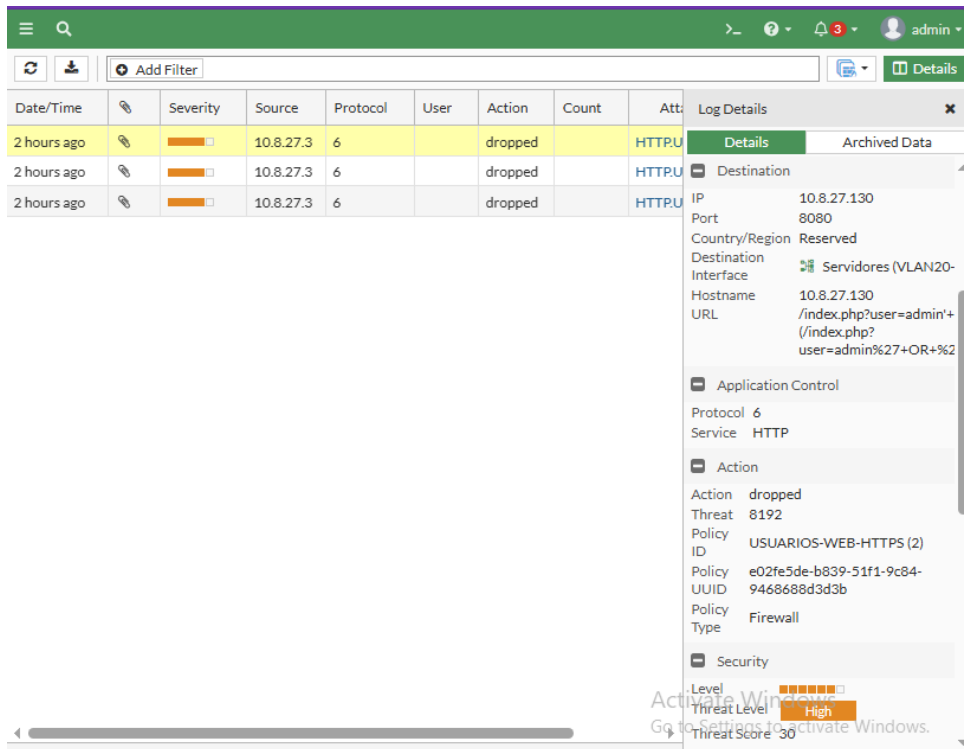
The screenshot shows the FortiGate configuration page for 'Edit IPS Sensor'. The left sidebar contains navigation options: FortiGate-VM64-KVM, Dashboard, Network, Policy & Objects, Security Profiles (selected), AntiVirus, Web Filter, Video Filter, DNS Filter, Application Control, Intrusion Prevention (selected), File Filter, SSL/SSH Inspection, Application Signatures, IPS Signatures, Web Rating Overrides, Web Profile Overrides, and VPN.

The main configuration area is titled 'Edit IPS Sensor'. It includes fields for Name (SQL-INJECTION-PROTECTION), Comments (Write a comment...), and Block malicious URLs (checked). Below this is the 'IPS Signatures and Filters' section, which contains a table with columns: Details, Exempt IPs, Action, and Packet Logging.

Details	Exempt IPs	Action	Packet Logging
HTTP.URI.SQL.Injection	0	Block	Enabled

At the bottom, there is a 'Botnet C&C' section with a 'Scan Outgoing Connections to Botnet Sites' option, which has buttons for 'Disable', 'Block', and 'Monitor'.

Prueba 17 — Detección y bloqueo de intento de SQL Injection



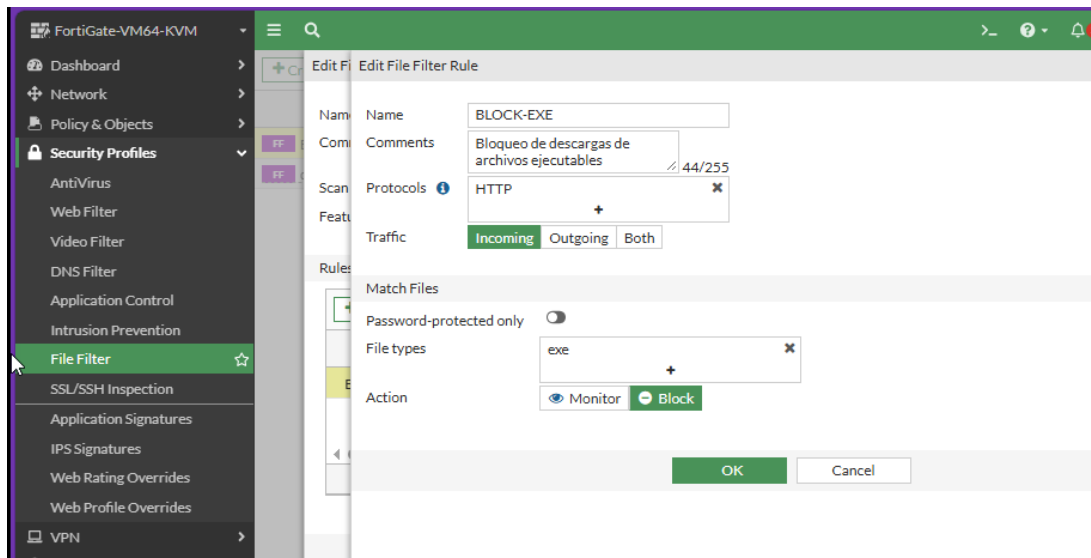
The screenshot displays the FortiGate log interface. The main table shows three log entries, all marked as 'dropped' and originating from 10.8.27.3. The 'Log Details' panel on the right provides a comprehensive breakdown of the event.

Date/Time	Severity	Source	Protocol	User	Action	Count	Att
2 hours ago	[Orange icon]	10.8.27.3	6		dropped		HTTPU
2 hours ago	[Orange icon]	10.8.27.3	6		dropped		HTTPU
2 hours ago	[Orange icon]	10.8.27.3	6		dropped		HTTPU

Log Details:

- Destination:** IP: 10.8.27.130, Port: 8080, Country/Region: Reserved, Destination Interface: Servidores (VLAN20-), Hostname: 10.8.27.130, URL: /index.php?user=admin'+/index.php?user=admin%27+OR+%2
- Application Control:** Protocol: 6, Service: HTTP
- Action:** Action: dropped, Threat: 8192, Policy ID: USUARIOS-WEB-HTTPS (2), Policy UUID: e02fe5de-b839-51f1-9c84-9468688d3d3b, Policy Type: Firewall
- Security:** Level: [Orange bar], Threat Level: High, Threat Score: 30

Prueba 18 — Configuración de File Filter para bloquear archivos .exe



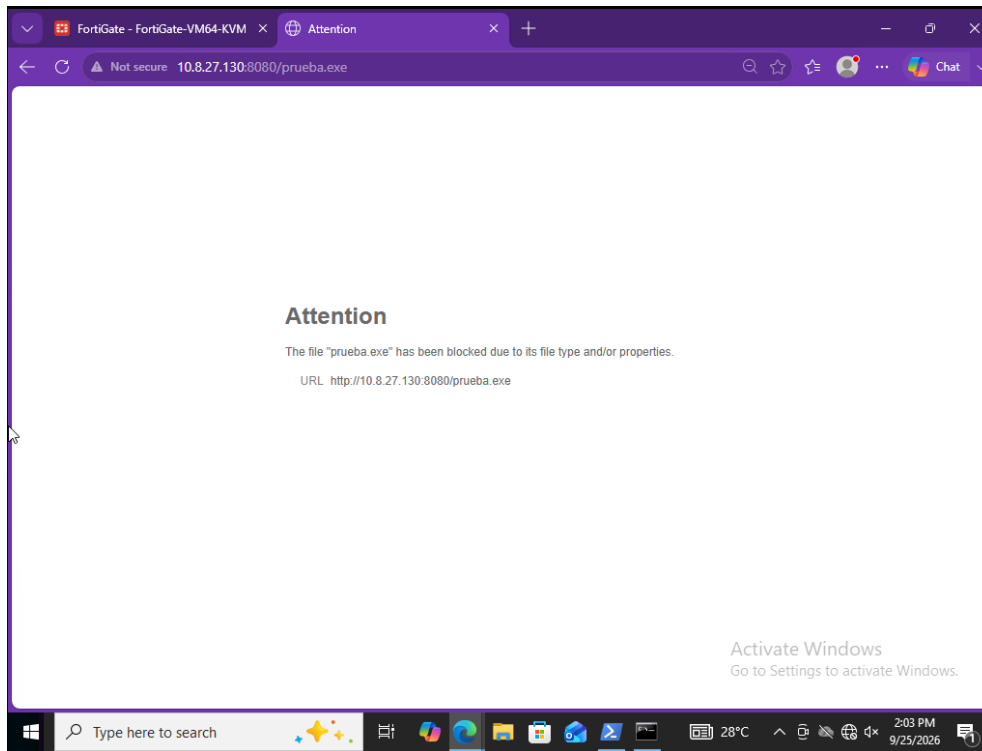
The screenshot shows the 'Edit File Filter Rule' configuration window in FortiGate. The 'File Filter' option is selected in the left-hand menu. The configuration is set to block executable files (.exe) over HTTP.

Edit File Filter Rule

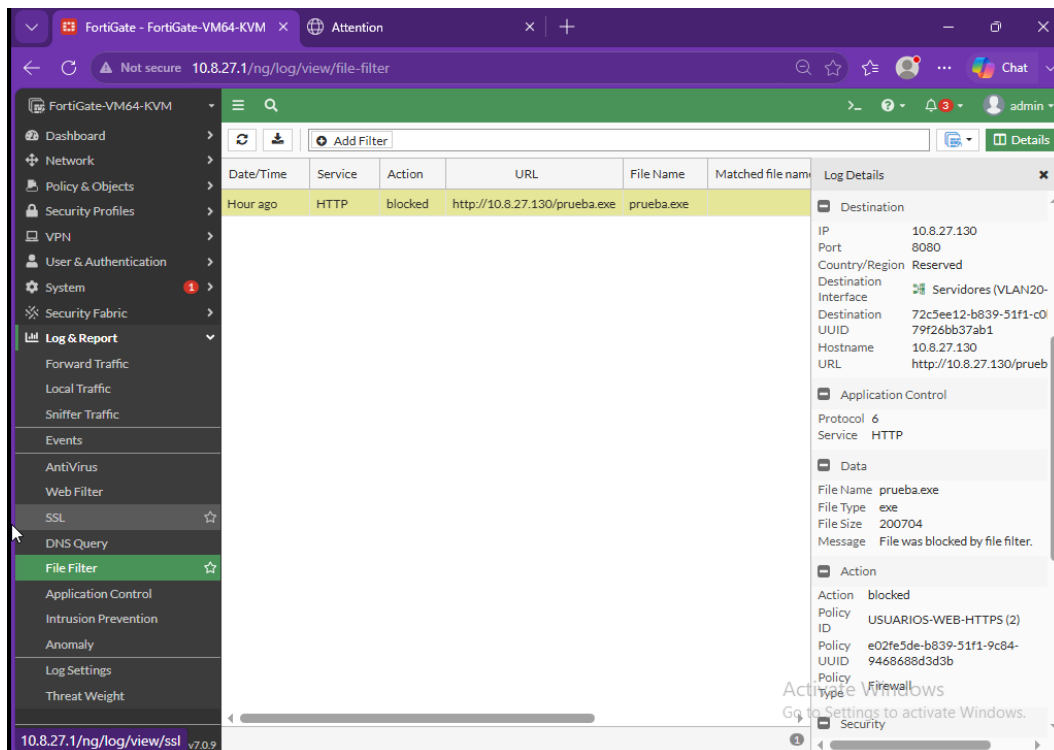
- Name:** BLOCK-EXE
- Comments:** Bloqueo de descargas de archivos ejecutables
- Scan Protocols:** HTTP
- Traffic:** Incoming, Outgoing, Both (Incoming is selected)
- Match Files:** Password-protected only (disabled), File types: exe
- Action:** Monitor, Block (Block is selected)

Buttons: OK, Cancel

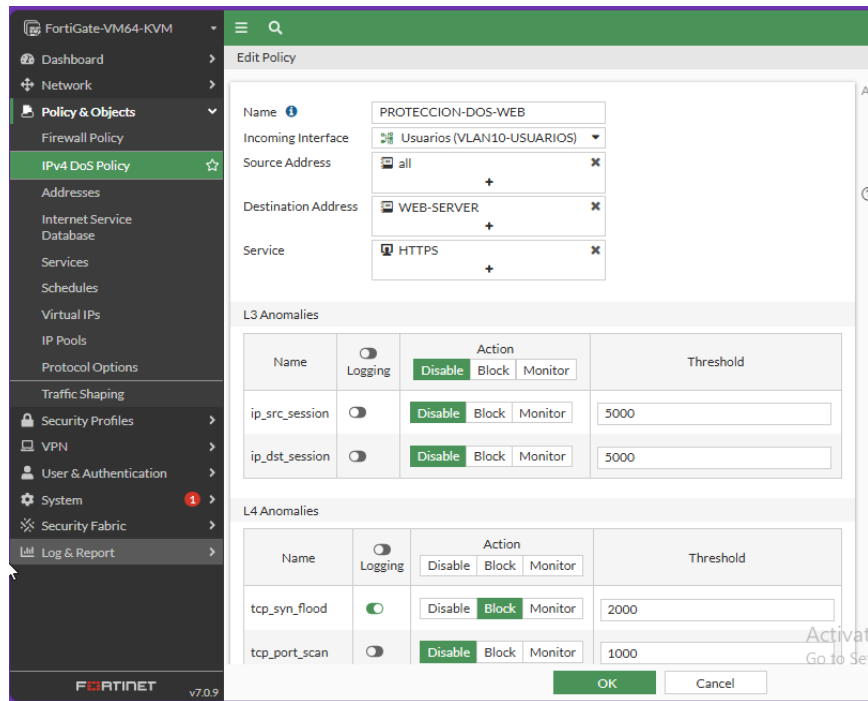
Prueba 19 — Bloqueo de descarga del archivo prueba.exe



Prueba 20 — Registro del bloqueo del archivo prueba.exe en File Filter



Prueba 21 — Configuración de protección DoS para el WEB Server



Prueba 22 — Verificación de las VLANs configuradas en el switch

```
Switch>en
Switch#show vlan brief

VLAN Name                                Status    Ports
----
1    default                                active    Gi1/0, Gi1/1, Gi1/2, Gi1/3
                                           Gi2/0, Gi2/1, Gi2/2, Gi2/3
                                           Gi3/0, Gi3/1, Gi3/2, Gi3/3
10   USUARIOS                                active    Gi0/1
20   SERVIDORES                             active    Gi0/2, Gi0/3
1002 fddi-default                         act/unsup
1003 token-ring-default                   act/unsup
1004 fddinet-default                       act/unsup
1005 trnet-default                         act/unsup
Switch#
```

Prueba 23 — Verificación del enlace trunk entre el switch y FortiGate

```
Switch#show interfaces trunk

Port      Mode      Encapsulation  Status      Native vlan
Gi0/0     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Gi0/0     10,20

Port      Vlans allowed and active in management domain
Gi0/0     10,20

Port      Vlans in spanning tree forwarding state and not pruned
Gi0/0     10,20
Switch#
```

Prueba 24 — Verificación de Port Security en el switch

```
Switch#show port-security
Secure Port  MaxSecureAddr  CurrentAddr  SecurityViolation  Security Action
          (Count)        (Count)          (Count)
-----
      Gi0/1           2             1              0             Shutdown
      Gi0/2           2             1              0             Shutdown
      Gi0/3           2             1              0             Shutdown
-----
Total Addresses in System (excluding one mac per port)    : 0
Max Addresses limit in System (excluding one mac per port) : 4096
Switch#
```

Prueba 25 — Verificación de puertos no utilizados deshabilitados en el switch

```
Switch#show interfaces status

Port      Name                Status      Vlan    Duplex  Speed  Type
Gi0/0     TRUNK-A-FORTIGATE   connected   trunk   a-full  auto   RJ45
Gi0/1     PC1-VLAN10          connected   10      a-full  auto   RJ45
Gi0/2     WEB-SERVER-VLAN20   connected   20      a-full  auto   RJ45
Gi0/3     DB-SERVER-VLAN20    connected   20      a-full  auto   RJ45
Gi1/0                      disabled    1        auto    auto    RJ45
Gi1/1                      disabled    1        auto    auto    RJ45
Gi1/2                      disabled    1        auto    auto    RJ45
Gi1/3                      disabled    1        auto    auto    RJ45
Gi2/0                      disabled    1        auto    auto    RJ45
Gi2/1                      disabled    1        auto    auto    RJ45
Gi2/2                      disabled    1        auto    auto    RJ45
Gi2/3                      disabled    1        auto    auto    RJ45
Gi3/0                      disabled    1        auto    auto    RJ45
Gi3/1                      disabled    1        auto    auto    RJ45
Gi3/2                      disabled    1        auto    auto    RJ45
Gi3/3                      disabled    1        auto    auto    RJ45
Switch#
```